ECON 100B Spring 2026 Your name and SID:

Problem Set 2

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

Did you work with other students? List them below:

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

> "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others."

Type your name:

1. [18 points total] Short questions with answers so low and a little Roman.

(a) [2 points] Recall that real GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \tag{1.1}$$

What happens if there is no government and no foreign trade? Rewrite equation (1.1) and briefly explain what it states about the disposition of income, Y. Does a component of income become interpretable as something else? Explain.

(b) [2 points] In the U.S., what can you say about the level of the long-run share of consumption in GDP, meaning C/Y? Briefly discuss.

(c) [2 points] Consumption goods and services include things like food and health care. Investment goods and services include container cranes for seaports, architectural services, and the office buildings themselves. Do the same workers in the economy typically produce both consumption goods and investment goods? Explain.

(d) [2 points] Given your previous answers, what should be true in the short run about the U.S. share of labor engaged in producing consumption goods, if consumption and investment sectors require and benefit from the same amount of capital and technology?

(e) [2 points] Recall that when production is Cobb-Douglas with a capital share $\alpha = 1/3$, the marginal product of capital is given by:

$$MPK = \frac{1}{3}\bar{A}\left(\frac{L}{K}\right)^{2/3} \quad (1.2)$$

Suppose we knew these calibrated[1] values for A, K, and L:

A = 1,983	"Total Factor Productivity" is a residual
K = 80,823,096	The U.S. capital stock is about $80 trillion
L = 163	U.S. employment is about 163 million

Using a calculator or a spreadsheet, insert these values into the right-hand side of equation (1.2) and report and comment on what you calculated. In words, what does your value mean? Round to the nearest thousandth = tenth of a percent and interpret.

(f) [2 points] Imagine that employment is reduced by 10 million workers to 153 million. Assuming nothing else changes, repeat your calculation above and report the MPK. What has happened? Why? Express the change in level and in percentage terms.

[1] The Penn World Table 11.0 (2026) reports current-price capital stock in millions of PPP adjusted 2021 US$ (variable "cn") as $80,823,096 in 2023, with expenditure side GDP ("cgdpe") as $25,586,132 million, for an aggregate K/Y = 3.16. In early 2026, BLS reported 163 million employed. Solow model calibration by Matthew Shapiro's comment in Eric M. Engen & R. Glenn Hubbard (2005) "Federal Government Debt and Interest Rates," *NBER Macroeconomics Annual 2004, (19): 83-160* is gratefully acknowledged.

(g) [2 points] Recall that GDP is given by:

$$Y = A K^{1/3} L^{2/3} \qquad (1.3)$$

Use the calibrated values above (on the previous page) to calculate Y and Y' with A and K, and $L = 163$ and $L' = 153$. Report those levels of GDP and describe what you have found. What is the percentage change in GDP caused by this reduction in labor? Also, what was the percentage reduction in labor?

(h) [2 points] Take the growth rate of both sides of equation (1.2) and briefly explain what you see.

(i) [2 points] Are your answers to parts (f) and (g) and (h) consistent? Use your answer to (h) to predict what the percentage change in MPK and compare it to what you calculated. Explain.

2. [40 points total] Immigration and capital remittances in the Solow Model.

The saving rate plays a key role in the Solow Model. In particular, when the economy is closed and there are no cross-border exchanges, saving directly fuels domestic investment and increases the domestic capital stock. But in a world with mobile labor and capital, what are the implications of cross-border flows for well-being?

Imagine the U.S. is a Solow economy, and consider the effects of a one-time flow of immigration. Recall that when labor is not growing, the key equations of the Solow Model are:

$$Y = A\,K^{1/3}\,L^{2/3} \qquad (2.1)$$

$$\Delta K = s\,A\,K^{1/3}\,L^{2/3} - dK \qquad (2.2)$$

(a) [4 points] On the axes below, draw the Solow diagram identifying the saving, depreciation, and income curves, and the steady state. Briefly explain what you have drawn.

(b) [6 points]Now redraw the Solow diagram showing the effect(s), if any, of a one-time increase in the labor supply, L, due to a substantial amount of immigration. Assume nothing else changes, and explain, discussing short and long-run effects.

So far, we have assumed that immigrants save at the same rate as natives, which seems reasonable. But what would happen if immigrants sent some of their saving back to their native countries, where it became foreign capital?

(c) [4 points] Redraw part (b) showing the full effects of immigration: the increase in **L** and any change in the national investment rate **s** owing to immigration. On the next page, explain what you have drawn, discussing short and long-run effects.

(d) [2 points] In the steady state, what is true about ΔK? Rewrite equation (2.2) using this.

(e) [2 points] Substitute equation (2.1) into your answer to part (d) to find an equation that includes A, K, L, s, and d.

(f) [2 points] Manipulate your answer to part (e) to find an equation for the steady-state level of capital, K*, in terms of A, L, s, and d. Show your work.

(g) [2 points] Take the growth rate of both sides of your answer to part (f) to find a formula for the growth rate of K in terms of the growth rates of A, L, s, and d. Simplify.

(h) [2 points] Take the growth rate of both sides of equation (1) to find a formula for the growth rate of output, Y , in terms of the growth rates of A, K, and L. Simplify.

(i) [2 points] (i) (2 points) Finally, subtract the growth rate of L from your answer to part (h) to find a formula for the growth rate of output per person, Y/L, in terms of the growth rates of A, K, and L. Simplify.

Today, the share of Americans who are foreign-born is around 15-16 percent. For simplicity, let us assume that statistic is 10 percent, and although immigrants actually arrive in much smaller numbers, let us examine the effects of a one-time increase in L equal to 10 percent.

(j) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the short run when L rises by 10 percent? Use your answer to part (i) and be specific.

(k) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the long run when L rises by 10 percent? Use your answers to parts (g) and (i) and be specific.

Now let us think about potential effects of immigration on saving and capital. Assume the original saving rate **s** among U.S. natives was 20 percent, and let us take an extreme view: that all immigrants send all of their saving abroad. Thus their contribution to the national investment rate is zero.

(l) [3 points] If 90 percent of residents save at 20 percent, and 10 percent of residents save at 0 percent, what is the new national or average saving rate, or is it the same? (Hint: this is like reducing the number of savers by the percent who are immigrants.)

(m) [5 points] What are the long-run effects of immigration on output per person if the saving rate does as you have described in part (l)? Use your answers to parts (g) and (i) and be specific.

3. [18 points total] The Romer Model and the Zombie Apocalypse.

Braaaaaaaains! Consider an "old school" Romer economy[2] like the one we saw in class, with production of knowledge A and food Y. Scientists produce new knowledge using old knowledge and have productivity $\bar{z}$. Total population is fixed at $\bar{N}$ and employed either in knowledge or in food. The share of population in knowledge production equals ℓ, a number like 0.10.

$$\Delta A_t = \bar{z}\, A_t\, L_{at} \tag{3.1}$$

$$Y_t = A_t\, L_{yt} \tag{3.2}$$

$$\bar{N} = L_{at} + L_{yt} \tag{3.3}$$

$$L_{at} = \ell\, \bar{N} \tag{3.4}$$

Recall that the solution to the Romer Model reveals the following:

$$g[A] = \bar{z}\, \ell\, \bar{N} \tag{3.5}$$

$$A_t = A_0\,(1 + g[A])^t \tag{3.6}$$

$$y_t = A_0\,(1 + g[A])^t (1 - \ell) \tag{3.7}$$

Imagine what might happen if all the scientists accidentally infect themselves with the "zombie flu" and become zombies. Zombies:

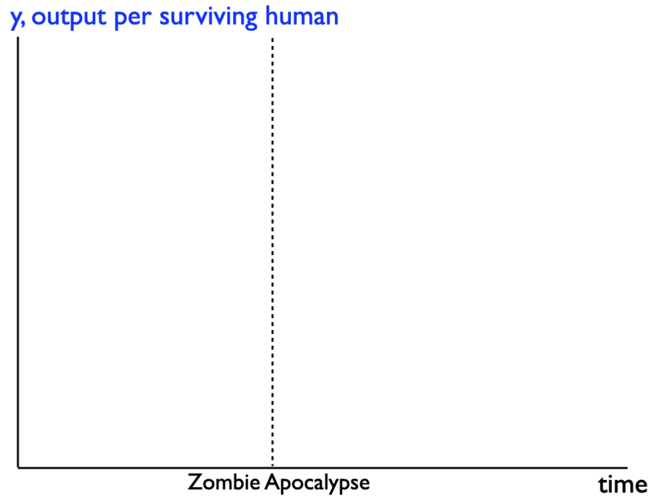
- Do not produce new ideas
- Do not destroy old ideas
- Eat brains and do not eat food produced by farmers
- Do not eat farmers' brains. They prefer tasty imported brains

(a) [3 points] When the scientists become zombies, what happens to the growth rate of A? What about the growth rate of y?

(b) [3 points] When the scientists become zombies, what happens to the level of A? What about the level of output per person? (Consult the bullet points about zombies above.)

[2] The Romer model in Jones 6e is a little different. The "old school" Romer model is what appears in editions 1-5 and in the course slides for ECON 100B.

(c) [4 points] Illustrate the Zombie Apocalypse in the graph below, showing any changes in the levels and growth rates in output per person over time. Discuss what you have drawn. Is the zombie flu good or bad? Explain.



(d) [4 points] Suppose rather than being turned into zombies, the scientists had all been deported to another country by the government. Would anything be different? Why?

(e) [4 points] Suppose rather than being turned into zombies, the scientists had all been repurposed as farmers by the government. Would anything be different? Why?